

Computational Behavior Science:

Quantifying Human and Animal Behavior using Python programming, data science, and applied AI/ML

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INTRODUCTION

From the precise movements of an athlete or a dancer to the intricate decision-making in animals, humans, and organizations—each represents a form of behavior at a different scale. Therefore, understanding and predicting behavior is fundamental to advancing research and driving innovation across disciplines. Precise, objective measurement of behavior forms the foundation of such research. For such quantification, behavior is defined as a temporal sequence of discrete actions—whether movements, vocalizations, or other measurable signals—occurring in a given context.

These quantitative measurements underpin advances in neuroscience, medical sciences, biomechanics, biophysics, biomimetics, and ecology, as well as transformative applications in sports, performing arts, and digital media.

This module gives students the opportunity to investigate behavior in humans and other animals using modern computational and AI/ML-based tools. Students may design and execute a project focused on:

- **Quantifying movement** from video using computer vision and pose estimation.
- **Quantifying audio signals** such as vocalizations or musical performance.
- **Combining multimodal data** (digital records, video and audio) to create a richer behavioral profile.

Projects are conducted in a collaborative research environment that mirrors real-world scientific practice, with opportunities for cross-project feedback and knowledge sharing.

GOAL OF THE COURSE: WHAT WILL YOU LEARN?

Beyond applying AI to behavioral science, students will gain a practical understanding of the full research cycle—from identifying a compelling question to presenting findings to diverse audiences.

Specifically, you will learn to:

- Formulate a behavior-related questions as testable scientific hypotheses.
- Utilize and fine-tune open-source machine learning models to extract and analyze behavioral metrics from various types of data – digital data records, audios, videos.
- Apply data science and perform statistical analyses in Python to validate results.
- Incorporate and AI/ML techniques, such as computer vision,
- Responsible and tactful integration of large language models in research.
- Communicate your work effectively through oral presentations, written research reports, and visualizations.

SUGGESTED PROJECTS

While you are encouraged to create your own research questions, below are some examples of techniques you can use to answer your questions:

PROJECT TYPE 1: Movement Analytics for Sports, Dance & Rehabilitation

Record practice or rehabilitation sessions and use computer vision-based pose tracking to quantify performance. Analyze posture dynamics, identify patterns linked to efficiency or risk of injury, and develop feedback systems for skill improvement.

PROJECT TYPE 2: Tracking Vocal or Instrument Learning

Record practice sessions for singing or playing a musical instrument, extract audio features (e.g., pitch stability, timing accuracy), and analyze progress over time using AI/ML-assisted scoring and classification techniques.

PROJECT TYPE 3: Behavior Prediction from Video

Select an animal group, gather ~50 videos (~200 minutes), extract poses, classify distinct behaviors, and use predictive models (e.g., temporal graph neural networks) to forecast future behavior in novel videos.

REQUIREMENTS

- Computer (Mac/Windows/Linux) with at least 8 GB RAM and internet connection.
- A Google account.
- Access to video or audio data (if needed, required recording equipment will be provided).
- No prior programming experience required, but familiarity with Python is a plus.

EXPECTATIONS FROM STUDENTS

- Work in teams of 2–4 members but submit assignments individually on time.
- Estimated time commitment: 6–10 hours/week.
- Attend all weekly meetings, having reviewed assigned materials in advance.
- Present at least one relevant research paper during the semester.
- Participate actively in discussions, presentations, and peer feedback sessions.

WHO SHOULD TAKE THIS COURSE?

This course will be ideal for students interested in neuroscience, psychology, biology, computer science, sports science, or the arts who wish to explore application of computation, data science and AI/ML workflows in the study of movement and behavior across species.